

# CS 4530: Fundamentals of Software Engineering

## Module 1.2: User Stories

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# Learning Goals for this Lesson

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- At the end of this lesson, you should be able to
  - Explain the overall purposes of requirements analysis
  - Explain the connection between requirements analysis and user stories
  - Explain the structure of a user story
  - Identify and fix user stories that don't have the correct structure
  - Define the relationship between conditions of satisfaction and user stories, and the difference between essential, desired, and extension conditions of satisfaction
  - Explain the difference between functional and non-functional requirements, and give examples of each

# Requirement analysis helps in making sure that we are building the right thing

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**How the customer explained it.**



**How the project leader understood it.**



**How the analyst designed it.**



**How the programmer wrote it.**



**What the customer really wanted.**

# Why is requirements analysis hard?



## Problems of understanding

Do users know what they want?  
Do users know what we don't know?  
Do we know who are users even are?



How the customer explained it.



How the project leader understood it.



## Problems of scope

What are we building?  
What non-functional quality attributes are included?



What the customer really wanted.



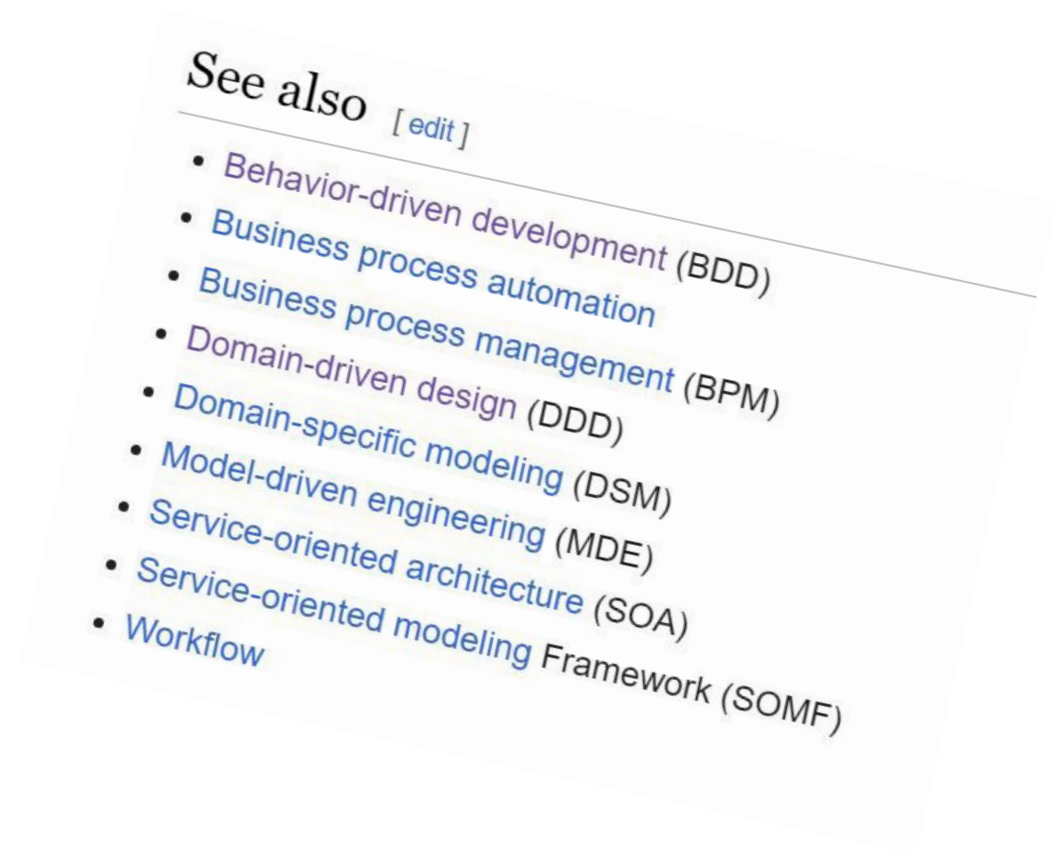
## Problems of volatility

Changing requirements over time

# How do we capture the requirements?

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- There are many frameworks for capturing requirements.
- Often described as  $x$ -Driven Design (for some  $x$ )
- They differ in scope & details, but they have many features in common.



# Common Elements

- Meet with stakeholders
- Develop a common language
- Collect desired system behaviors that offer value
- Document the desired behaviors
- Iterate and refine!!

**User stories** are the least common denominator of most approaches

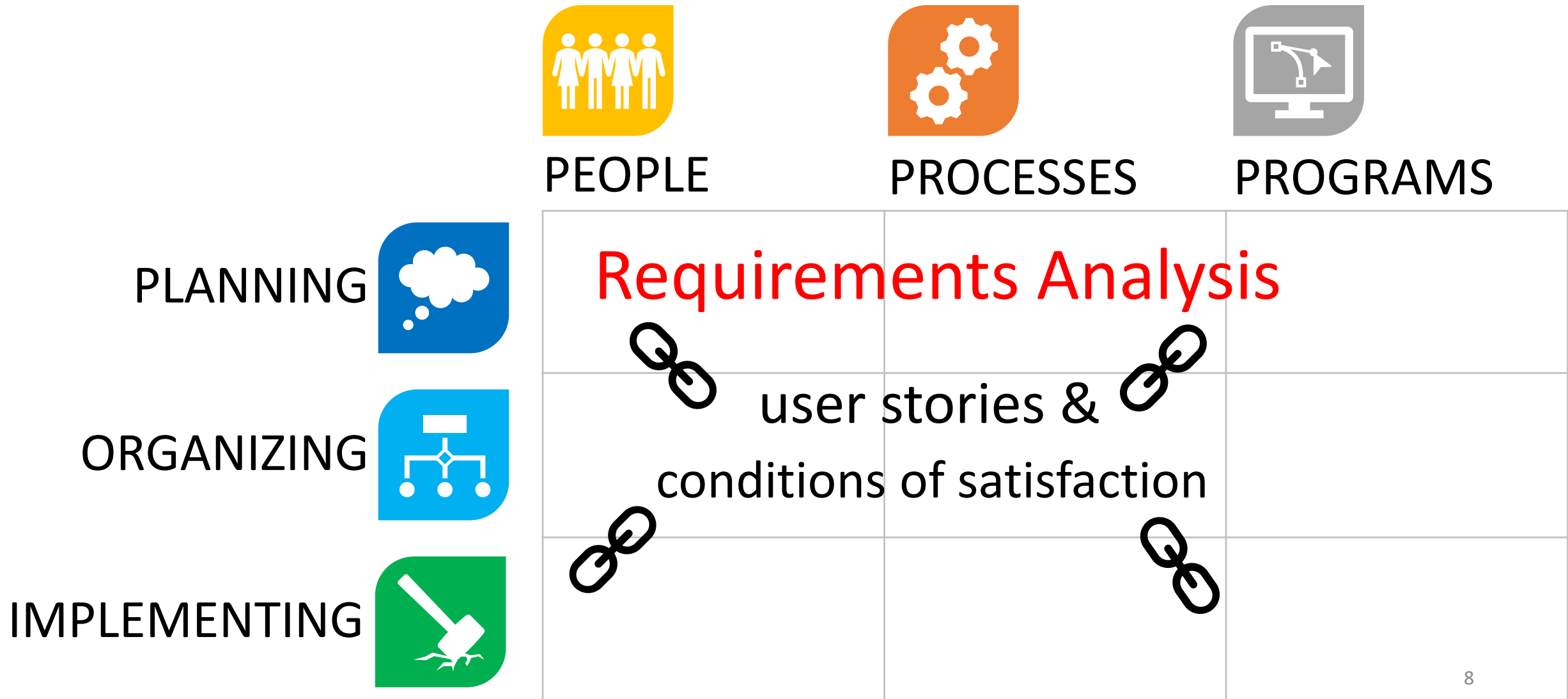


# “Building the Right Thing” Necessarily involves Value Judgments

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- Right for whom?
- Who benefits?
- Limitations of budget/time/personnel
- Competing/incompatible ideas about what's important
- Competing/incompatible sets of Conditions of satisfactions (COSs)

# Requirements Analysis connects High-Level Planning To User Stories





# User stories come from analyzing the user's requirements

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*As a(n) active outdoor park-goer  
I want a safe way to fly through the air under a tree  
so that I can feel the wind in my hair*

# What are user stories?

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- ...a least-common-denominator approach documenting the requirements of users when **planning** projects
- ...a tool for keeping large collaborative teams on the same page when **organizing** projects
- ...formulaic statements of this form:

*As a <role>*

*I want <capability>*

*so that I can <get some benefit>*

# Components of a user story

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*As a(n) active outdoor park-goer  
I want a safe way to fly through the air under a tree  
so that I can feel the wind in my hair*

*As a <role>  
I want <capability>  
so that I can <get some benefit>*

# Roles: “who”

*Think “Canvas roles”*

- Roles are positions or functions that people inhabit! The person is not you! It is the user, not customer
  - You may want to build software to develop skills, or to make a codebase more maintainable. That’s good, but it doesn’t fit as a user story.
- User stories are for users in a specific role:
  - “As a web server...” is not a user story!
  - “As a human being...” is better, but that’s still not a role
  - “As a user...” is almost always a cop-out

*As a <role>  
I want <capability>  
so that I can <get some benefit>*

# Capabilities: “what”

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- A capability is a specific thing I want to be able to do
  - Because we’re building software, this is usually an action we can provide by using software, otherwise let’s not build software
- A capability relates to a role
  - “As a teacher at Northeastern, I want to be able to access a laser cutter so I can finish an art project” is not a good user story. (*It is too general*)

*As a <role>  
I want <capability>  
so that I can <get some benefit>*

# Capabilities: “what”

*Think verb, not noun*

- A capability is not a product
  - “As a College Administrator, I want a web application that does <this> and <that> so that I can...” is not a user story!
  - Better: “I want to be able to see student’s grades” or “I want a way to track course enrollment over time”
  - This is easy to get wrong in practice!  
Sometimes you really want to build a tic-tac-toe game.
  - You’re supposed to ask, “do I even need to build this?”

*As a <role>  
I want <capability>  
so that I can <get some benefit>*

# Benefit: “why?”

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- Benefits are key for user stories actually focusing on what matters to the user.
- If a specific feature doesn't relate to the benefit...
  - ...maybe that feature isn't worth building.
  - ...maybe that feature is part of a different user story. (Maybe we should prioritize that different user story instead?)

*As a <role>  
I want <capability>  
so that I can <get some benefit>*

# Properties of a good user story

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- short: fits on a 3x5 card
- may have prerequisites
- has *conditions of satisfaction* that expand on the details
- satisfies the INVEST+E criteria (more on this later)



# Good Examples of User Stories

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- As a College Administrator, I want to keep track of students, the courses they have taken, and the grades they received in those courses, so that I can advise them on their studies.
- As a driver, I want to be able to report potholes to the city so that the town can more quickly act to keep me safe.
- As a pedestrian, I want to be able to report potholes to the city so that drivers stop dangerously swerving onto the sidewalk when I'm walking.
- As a card game enthusiast, I want to be able to play blackjack online so that I can...

# Conditions of Satisfaction fill in details of the desired behavior

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- Each condition of satisfaction
  - Describes a testable behavior, from the user's point of view
  - Must have a priority
  - Should be numbered within its user story
- There may be *multiple* options for filling in the details with conditions of satisfaction: this is a design problem!



# Conditions of Satisfaction have Priorities. User stories too!

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- **Essential (E)** means the project is useless without it.
- **Desirable (D)** means the project is less usable without it, but it is still usable.
- **Extension (X)** describes a CoS that may not be achievable within the scope of the project. These might be things you'd want "in the next version".
- **Prioritizing user stories** is also a *planning activity* for *complex* projects that are constrained by resources (budget, time, personnel) and multiple (competing or incompatible) ideas about what's important.



# Worked Example: Pothole reporting system

A town is designing a system where citizens can report potholes and the town can monitor progress on repairing them.





# User Story #1

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- As a car commuter, I want to be able to report potholes to the city so that the town can more quickly act to keep me safe.

# Conditions of Satisfaction

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- As a car commuter, I want to be able to report potholes to the city so that the town can more quickly act to keep me safe.
  - 1.1 I should be able to report the location of a pothole to the system (E)
  - 1.2 I should be able to see whether the pothole I report has been repaired (E)
  - 1.3 I should be able to see whether others have reported potholes near me (D)
  - 1.4 I should be able to see an estimated time when the pothole should be repaired (X)

## User Story #2

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- As a pothole-repair-truck driver, I want the system to display the potholes I should be working on today, so that ... (?)

# Conditions of Satisfaction

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- As a pothole-repair-truck driver, I want the system to display the potholes I should be working on today, so that I can view and report my progress for the day.
  - 2.1 I should be able to see my list of potholes for today (E)
  - 2.2 I should be able to report that I repaired a given pothole (E)
  - 2.3 I should be able to report that I was unable to repair a given pothole, and to supply a reason (E)
  - 2.4 My daily list of potholes should be listed in an order that cuts down the time I spend driving from job to job (D)



## User Story #3

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- As a maintenance supervisor, I want to be able to control the order in which potholes are repaired, so that... (?)

# Conditions of Satisfaction

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- As a maintenance supervisor, I want to be able to control the order in which potholes are repaired, so that complaints to city hall are minimized.
  - 3.1 I should be able to give a higher priority to potholes on a particular street (E)
  - 3.2 I should be able to give a higher priority potholes in a particular neighborhood (E)
  - 3.3 Anyone not a maintenance supervisor **should not** be able to change pothole priority (E)
  - 3.4 I should be able to see on a map where there are a lot of potholes (D)
  - 3.5 I should be able to see on a map which potholes that have been reported multiple times (D)

# Writing User Stories: INVEST + E

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- Independent
- Negotiable
- Valuable (has value to client)
- Estimable (able to estimate development effort)
- Small
- Testable
- Ethical (connects with our & users' values)

*As a <role> I want  
<capability> so that I can  
<get some benefit>*

# User stories aren't everything!

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- User stories roughly describe the **functional requirements**
- The **non-functional requirements** are other properties that are also important to users and to other stakeholders
  - How quickly can a transcript be retrieval? (Performance)
  - How many student transcripts can our system store? (Scalability)
  - How long did I spend on the phone with support to set up the software? (Usability)
  - After my system is setup, is the access controlled at all? (Security)
  - Are there any times when I can't use this system? (Availability)
  - How expensive will it be to adapt the system to new requirements? (Maintainability)

# Other non-functional requirements

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- Availability
- Capacity
- Efficiency
- Extensibility
- Maintainability
- Performance
- **Privacy**
- Response Time
- Response Time
- Scalability
- **Security**
- Supportability
- Testability
- **Usability**

# Minimum Viable Product

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- Once you decide on a set of user stories, the Essential (E) conditions of satisfaction for those user stories constitute the minimum viable product (MVP)
- A user story is "implemented " when all its essential COSs are implemented.
- Caution: when proposing a project, don't make your MVP too hard to complete (but don't make it too easy, either)
- For course project, MVP is worth 8% of the course grade and desirable features are worth additional 4%.

# Review

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